



PHISHING EMAIL DETECTION & AWARENESS REPORT

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Program : Future Interns – Cybersecurity Task 2

Date : May 29, 2026

Type : Email Security Analysis & Awareness

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WHAT IS PHISHING?

Phishing is a type of cyber attack where criminals send fake emails pretending to be trusted companies like banks, PayPal, or Microsoft.

Their goal is to trick users into:

- Clicking malicious links
- Entering passwords on fake websites
- Downloading infected files
- Sharing OTPs or personal information

WHY IT'S DANGEROUS:

- cause of data breaches worldwide
- Costs businesses billions annually
- Targets employees at all levels
- No technical skill needed to fall victim

REAL STATISTICS:

- 3.4 billion phishing emails sent daily
- 36% of data breaches involve phishing
- 97% of users cannot identify phishing emails
- Average cost per breach: \$4.9 million

TYPES OF PHISHING ATTACKS

1. EMAIL PHISHING

Mass emails sent to thousands of users

Example: Fake bank alerts, invoice scams

2. SPEAR PHISHING

Targeted attack on specific person/company

Uses personal information to seem legitimate

3. SMISHING (SMS Phishing)

Fake text messages with malicious links

Example: "Your package is held, click here"

4. VISHING (Voice Phishing)

Fake phone calls pretending to be support

Example: "Your computer has a virus"

5. WHALING

Targets senior executives (CEO, CFO)

High-value targets for maximum impact

6. CLONE PHISHING

Copies legitimate email, replaces links

Looks identical to real company emails

7. PHARMING

Redirects users to fake websites

Even correct URL leads to phishing site



EMAIL SAMPLE 1 — FAKE BANK OF AMERICA ALERT

RISK LEVEL: ● PHISHING CONFIRMED

EMAIL DETAILS:

From : security@bankofamerica-alert.com
Subject : Urgent: Your Account Has Been Suspended
Target : Banking credentials

HEADER ANALYSIS RESULTS:

SPF : FAIL — Unauthorized server
DKIM : FAIL — Not digitally signed
DMARC : FAIL — Domain policy violated
Tool : PhishKit v2.3 detected

PHISHING INDICATORS FOUND:

- Fake domain (bankofamerica-alert.com)
Real domain: bankofamerica.com
- Urgency language: "Account Suspended"
- 24-hour deadline threat
- Generic greeting: "Dear Valued Customer"
- Suspicious verification link
- Fear-based content

ATTACK TYPE: Credential Harvesting

GOAL: Steal bankinglogin credentials

Google Admin Toolbox Messageheader

MessageId20260528102339.x12si@bankofamerica-alert.com

Created at5/28/2026, 10:53:39 PM GMT+5:30 (Delivered after 2 sec)

From"Bank of America Security" <security@bankofamerica-alert.com> Using PhishKit v2.3

Touser@gmail.com

SubjectUrgent: Your Account Has Been Suspended

SPFfail with IP Unknown! [Learn more](#)

DKIMfail with domain Unknown! [Learn more](#)

DMARCfail [Learn more](#)

#	Delay	From *	To *	Protocol	Time received
0	2 sec	bankofamerica-alert.com	→ [Google] mx.google.com	ESMTP	5/28/2026, 10:53:41 PM GMT+5:30

ANALYZE ANOTHER HEADER

SHOW RAW HEADER

Email Sample 2 — Fake PayPal Invoice

Risk Level: ● Phishing Confirmed

Email Details:

From : service@paypal-invoice-notice.com
Subject : Invoice #INV-2026-4821 Payment Required
Target : Financial fraud / payment theft

Header Analysis Results:

SPF : FAIL — Unauthorized server
DKIM : FAIL — Not digitally signed
DMARC : FAIL — Domain policy violated
Tool : PhishKit v2.3 detected

Whois Result:

Domain paypal-invoice-notice.com = NOT REGISTERED
Completely fake domain created for fraud

Phishing Indicators Found:

- Fake unregistered domain
- Fake invoice amount (\$299.99)
- 12-hour payment threat
- Fake dispute link
- Fake phone number
- Automatic deduction threat

Attack Type: Financial Fraud

Goal: Steal money or payment credentials

Google Admin Toolbox Messageheader

Help

MessageId	20260528111522.paypal@invoice-notice.com
Created at:	5/28/2026, 11:45:22 PM GMT+5:30 (Delivered after)
From:	"PayPal Services" <service@paypal-invoice-notice.com> Using PhishKit v2.3
To:	user@gmail.com
Subject:	Invoice #INV-2026-4821 Payment Required
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0		paypal-invoice-notice.com	→ [Google] mx.google.com	ESMTP	5/28/2026, 11:45:22 PM GMT+5:30

ANALYZE ANOTHER HEADER

SHOW RAW HEADER

Whois

Domains Hosting Websites* Email Security Whois Deals

Enter Domain or IP

WHOIS

paypal-invoice-notice.com is available!

Sale

.space

~~\$29.88~~ \$1.18

BUY NOW

*while stocks last

On Sale!

.CLUB

.CLUB @ \$2.28 ~~\$24.48~~

AI WEBSITE BUILDER

Build A Beautiful Website In Minutes!

Starts at \$2.48

Email Sample 3 — Fake Microsoft 365

RISK LEVEL: ● PHISHING CONFIRMED

Email Details:

From : admin@microsoft-365-renewal.com
Subject : Action Required: Microsoft 365 Expiring
Target : Corporate login credentials

Header Analysis Results:

SPF : FAIL — Unauthorized server
DKIM : FAIL — Not digitally signed
DMARC : FAIL — Domain policy violated
Tool : PhishKit v2.3 detected

Whois Result:

Domain microsoft-365-renewal.com = NOT REGISTERED
Completely fake domain

Phishing Indicators Found:

- Fake unregistered domain
- 24-hour license expiry threat
- Credential stealing link
- Generic greeting: "Dear User"
- Fear of losing files and emails
- URL doesn't match microsoft.com

ATTACK TYPE: Corporate Credential Harvesting
GOAL: Access company Microsoft accounts

Google Admin Toolbox MessageheaderHelp

MessageId	20260528091013.ms365@renewal.com
Created at:	5/28/2026, 9:40:13 PM GMT+5:30 (Delivered after 2 sec)
From:	"Microsoft Support" <admin@microsoft-365-renewal.com> Using PhishKit v2.3
To:	employee@company.com
Subject:	Action Required: Microsoft 365 License Expiring
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0	2 sec	microsoft-365-renewal.com	→ [Google] mx.google.com	ESMTP	5/28/2026, 9:40:15 PM GMT+5:30

ANALYZE ANOTHER HEADER

SHOW RAW HEADER

WhoisDomainsHostingWebsites*EmailSecurityWhoisDeals

Enter Domain or IPWHOIS

Looks like this domain has not been registered yet

microsoft-365-renewal.com

\$13.48

BUY NOW

Before someone else does!

.space

~~\$29.88~~ \$1.18

BUY NOW

*while stocks last

On Sale!

fun

.FUN @ \$1.48 ~~\$42.88~~

AI+ WEBSITE BUILDER

Build A Beautiful Website In Minutes!

Starts at \$2.48

PHISHING INDICATORS CHECKLIST

ALWAYS CHECK THESE:

SENDER DOMAIN

- Does domain match real company exactly?
- Any extra words? (alert, secure, notice)
- Different extension? (.net instead of .com)

LINKS & URLs

- Hover before clicking — does it match?
- Is it HTTP or HTTPS?
- Does domain match the company?
- Any random numbers or words in URL?

EMAIL CONTENT

- Generic greeting? (Dear User/Customer)
- Urgency/threat language?
- Grammar or spelling errors?
- Unexpected attachment?

TECHNICAL CHECKS

- SPF: Pass or Fail?
- DKIM: Pass or Fail?
- DMARC: Pass or Fail?

RISK LEVELS

- **SAFE** → All checks pass
- **SUSPICIOUS** → Some checks fail
- **PHISHING** → Multiple checks fail

Whois

Identify for everyone

Domains

Hosting

Websites

Email

Security

Whois

Deals

Enter Domain or IP

WHOIS

bankofamerica-alert.com

Updated 1 second ago

Interested in similar domains?

Domain Information

Domain:

bankofamerica-alert.com

Registered On:

2022-01-16

Expires On:

2027-01-16

Updated On:

2026-01-12

Status:

client transfer prohibited

Name Servers:

ns101.bac.com
ns102.bac.com

Registrar Information

Registrar:

CSC Corporate Domains, Inc.

IANA ID:

299

Email:

admin@internationaladmin.com

Abuse Email:

domainabuse@cscglobal.com

Abuse Phone:

8887802723

Registrant Contact

Organization:

Bank of America Corporation

State:

NC

Country:

US

Email:

https://domaincontact.cscglobal.com/contactholder/bankofamerica-alert.com/registrant

bankofamericaalarm.com

Buy Now

bankofamericaalert24.com

Buy Now

bankofamericanotify.com

Buy Now

bankofamericanotification.com

Buy Now

bankofamerica-alert.net

Buy Now

bankofamericaalert.net

Buy Now

.space

\$29.88

\$1.18

BUY NOW

*while stocks last

On Sale!

fun

.FUN @ \$1.48 \$42.88

HOW A PHISHING ATTACK WORKS

STEP 1 → ATTACKER PREPARES

- Registers fake look-alike domain
- Creates clone of real website
- Sets up PhishKit email tool

STEP 2 → EMAIL SENT

- Mass email to thousands of targets
- Impersonates trusted brand
- Uses urgency and fear tactics

STEP 3 → USER CLICKS

- User panics due to urgency
- Clicks the malicious link
- Lands on fake website

STEP 4 → CREDENTIALS STOLEN

- User enters username/password
- Attacker captures credentials
- User sees "Thank you" page

STEP 5 → ATTACKER ACTS

- Logs into real account
- Steals money or data
- Sells credentials on dark web

STEP 6 → IMPACT

- Financial loss
- Identity theft
- Data breach
- Reputation damage

EMAIL RISK CLASSIFICATION TABLE

EMAIL SAMPLE	RISK	CLASSIFICATION
Bank of America Alert	HIGH	PHISHING CONFIRMED SPF/DKIM/DMARC FAIL
PayPal Invoice Notice	HIGH	PHISHING CONFIRMED FAKE UNREGISTERED DOMAIN
Microsoft 365 Renewal	HIGH	PHISHING CONFIRMED FAKE UNREGISTERED DOMAIN

CLASSIFICATION GUIDE :

- SAFE → SPF/DKIM/DMARC pass, known domain
- SUSPICIOUS → 1-2 checks fail, needs review
- PHISHING → Multiple failures, fake domain

PREVENTION GUIDELINES — DO'S
FOR ALL EMPLOYEES:

VERIFY THE SENDER

- Check exact domain spelling
- Compare with previous real emails
- Contact company directly if unsure

HOVER BEFORE YOU CLICK

- Check where link actually goes
- Look for HTTPS and correct domain
- Use VirusTotal to check suspicious URLs

USE MULTI-FACTOR AUTH (MFA)

- Even if credentials stolen, MFA protects
- Enable on all important accounts

REPORT SUSPICIOUS EMAILS

- Forward to IT security team immediately
- Don't delete — it helps investigation

KEEP SOFTWARE UPDATED

- Updated browsers block phishing sites
- Antivirus catches malicious downloads

VERIFY UNUSUAL REQUESTS

- Call the company using official number
- Never use contact info from the email



PREVENTION GUIDELINES – DON'TS

FOR ALL EMPLOYEES:

DON'T CLICK LINKS IN URGENT EMAILS

- Go directly to website by typing URL
- Urgency is a manipulation tactic

DON'T ENTER CREDENTIALS FROM EMAIL

- Always navigate to site independently
- Real companies never ask via email

DON'T OPEN UNEXPECTED ATTACHMENTS

- .zip, .exe, .doc from unknown senders
- Even PDFs can contain malware

DON'T CALL NUMBERS IN SUSPICIOUS EMAILS

- Numbers may connect to attacker
- Use official website for contact info

DON'T ACT ON FEAR OR URGENCY

- Stop and think before acting
- Verify before you trust

DON'T SHARE OTPs OR PASSWORDS

- No legitimate company asks via email
- This is always a scam

DON'T IGNORE BROWSER WARNINGS

- Red warning pages protect you
- Never click "proceed anyway"



EMPLOYEE AWARENESS TIPS

THE STOP-THINK-VERIFY METHOD:

STOP → Before clicking anything
Take 30 seconds to examine the email

THINK → Ask these questions:

- Was I expecting this email?
- Does the sender look right?
- Why is it so urgent?
- What happens if I ignore it?

VERIFY → Confirm legitimacy:

- Call the company directly
- Log in through official website
- Ask your IT team

QUICK EMAIL CHECK (30 SECONDS):

1. Look at sender email domain
2. Check for generic greeting
3. Look for urgency/threat language
4. Hover over any links
5. If unsure → report to IT

REMEMBER:

- Banks never email you to verify account
- PayPal never threatens auto-payment
- Microsoft never locks your account via email
- IT team never asks for your password
- If in doubt → throw it out!

CONCLUSION

This phishing email analysis identified and confirmed 3 phishing email samples targeting banking, payment, and corporate credentials.

KEY FINDINGS:

- All 3 emails failed SPF, DKIM, DMARC checks
- PhishKit v2.3 phishing tool detected in all samples
- 2 of 3 sender domains were not even registered
- All emails used fear and urgency tactics
- All impersonated trusted global brands

ANALYSIS TOOLS USED:

- Google Admin Toolbox — Header analysis
- WHOIS Lookup — Domain verification
- VirusTotal — URL safety check

RECOMMENDATIONS:

1. Train all employees on phishing awareness
2. Enable MFA on all corporate accounts
3. Deploy email security gateway (SPF/DKIM/DMARC)
4. Create clear phishing reporting procedure
5. Conduct regular phishing simulation tests

ETHICS STATEMENT:

All email samples used are for educational purposes only. No real attacks were performed.

Prepared By : Jagarlamudi Mokshagna

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THANK
YOU

